# The Cost of Transit Construction

*Stats 423: Applied Linear Regression*

March 11, 2026


| **Name** | **Contribution** |
|---|---|
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| Katie Gower | Introduction, Data Description |
| Viki Hristova | Abstract, Conclusion |



**Abstract**

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## 1 Introduction

Public transit works to move people back and forth from the important places in their lives every day. Many want the world at large to transfer to a more public transit oriented lifestyle, where less people use cars daily and more opt for public transport. When demand grows, it becomes necessary to add onto the existing transit infrastructure. However, cost of construction and the time to complete it are barriers in allowing this change.

The Transit Costs dataset was created with hopes of understanding what affects the cost of public transit construction projects. It covers many types of transit projects such as railroad, tunnel, elevated, lightrail, and subway and sixty countries across all continents except Antarctica. The observations are most concentrated in the past three decades.

### 1.1 Research Questions

1. What characteristics of projects drive cost per km?
2. What regions are projects more expensive in by kilometer?
3. What attributes create a more expensive project overall?
4. What regions are projects more expensive in overall?

# 2 Data Description and Exploratory Data Analysis (EDA)

## 2.1 Variables

Response variables:

- Cost per kilometer in millions of real US dollars
- Total real cost in millions US dollars

Predictors:

- Year: Midpoint of project completion
- Stations: Number of stations
- Length: Length of line in kilometers
- RR: Regional Rail or not (logical value)
- Country: Country Project is in
- Elevated: Length in kilometer of elevated line
- TunnelPer: Percentage of line that is tunnel
- AtGrade: Length of line atgrade (at ground level)
- Start Year
- End Year
- GDP: Real GDP for country and year

Variables created:

- Years: End Year - Start Year
- Continent: Continent the construction took place in

The Transit Costs dataset was created with hopes of understanding what affects the cost of public transit construction projects. It covers many types of transit projects such as railroad, tunnel, elevated, lightrail, and subway and sixty countries across all continents except Antarctica. The observations are most concentrated in the past three decades.

In the transit costs dataset, all monetary values are in millions of real US dollars. GDP was measured in real US dollars. Measures of length were in kilometers. We cast some columns as numeric or factor.

Some variables we were interested in weren't in the original dataset but could easily be created, like continent and years to complete.

The data had many missing values, but most of the integral ones were recovered. Real cost originally had most values missing, but when we multiplied real cost/kilometer and length in kilometers, it matched well to the existing values. We also added many values for

year using imputation. If a start year or end year was missing, it was filled in by taking the average years to complete for the continent of that observation. Asia and Africa originally had many more missing values for year than other continents.

GDP data was pulled from World Bank. The original format was rows corresponding to countries and columns corresponding to years. Data transformation was done to assign each row of the transit costs dataset a GDP value for the country of the project in the project's midpoint year. Urban population was also from World Bank and had a similar format, but did not make it into any of the final models.

### 2.2 Exploratory Analysis

Figure 1: Proportion of Missing Years by Continent

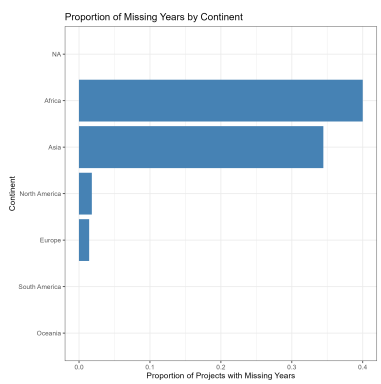


Figure 2: Proportion of Missing Years by Continent

## 3 Methodology and Model Building

Describe the models you fit to the data.

### 3.1 Proposed First Model

### 3.2 Proposed Second Model

We fit a multiple linear regression model of the form:

$$TotalCost = 57.91 + 0.157 * TunnelPer + 0.488 * Years + 1.223 * Length + -3.1e-13 * GDP - 31.23 * Continent$$

-45.75*ContinentEurope -24.36*ContinentNorthAmerica - 23.60*ContinentOceania-24.18*ContinentSou
+ $\epsilon_i(1)$

where $\epsilon_i \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$.

Explain how you selected your variables (e.g., stepwise selection, theoretical justification, handling of multicollinearity). If you used transformations (like taking the log of the response), explain why.

# 4 Results and Diagnostics

## 4.1 First Model Summary

This table summarizes the estimated coefficients for our first model:

| **Predictor** | **Estimate** | **Std. Error** | **t-value** | **Pr(> \|t\|)** |
|---|---|---|---|---|
| (Intercept) | -4.8386861 | 4.0279853 | -1.201 | 0.229993 |
| Years | 0.0274532 | 0.0060991 | 4.501 | 7.73e-06 |
| Start.year | 0.0054683 | 0.0020058 | 2.726 | 0.006542 |
| TunnelPer | 0.0055893 | 0.0004717 | 11.848 | $< 2e^{-16}$ |
| ContinentAsia | -1.3481124 | 0.2395711 | -5.627 | 2.51e-08 |
| ContinentEurope | -1.2433227 | 0.2384818 | -5.213 | 2.34e-07 |
| ContinentNorth America | -0.8059310 | 0.2467630 | -3.266 | 0.001136 |
| ContinentOceania | -1.0861986 | 0.3600030 | -3.017 | 0.002630 |
| ContinentSouth America | -0.9436826 | 0.2721792 | -3.467 | 0.000553 |
| PPP.rate | -0.1978282 | 0.0262669 | -7.531 | 1.32e-13 |

A log transformation created a more normal distribution of cost/km (insert histograms)

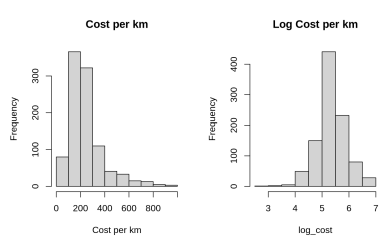


Figure 3: Histograms of Cost per kilometer Log Transformation

Initial model with as many variables: log_cost  Years + Start.year + Length + Stations + TunnelPer + PPP.rate + gdp + urban_pop

Bidirection stepwise selection, looking at AIC, only removed urban_pop.

Length and station are highly correlated (insert correlation matrix) so we removed the one that resulted in the best R squared.

Corr plot with log cost

GDP and PPP rate measure similar things and are slightly correlated. We removed the one that resulted in the smaller R squared, which was GDP.

Finally, we added continents as a categorical predictor to the model. After doing so, length became insignificant (high p-value) so we removed it, giving us the final model: (insert model summary - model name is cost_km_fin)

Costkm summary

Our first model focuses on cost per kilometer. This response variable was difficult to deal with and we ended up with an R squared value of 0.2252 after experimenting with different variables and transformations. This model had 824 degrees of freedom while there were 1007 original observations. The main insights of this model were than Asia and Europe were expected to have the cheapest construction costs when factoring in GDP separately, and that the amount of the line that was tunnel made the construction more expect, where if the entire line was tunnel, the model predicts it would cost five hundred sixty thousand dollars more than otherwise. However, since the R squared value is very low, it is hard to use this model for prediction.

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## 4.2 Second Model Summary

This table summarizes the estimated coefficients for our second model:

| **Predictor** | **Estimate** | **Std. Error** | **t-value** | **Pr($>\|t\|$)** |
|---|---|---|---|---|
| (Intercept) | 5.791e+01 | 8.365e+00 | 6.924 | 8.86e-12 |
| Years | 4.884e-01 | 2.064e-01 | 2.367 | 0.018175 |
| ContinentAsia | -3.123e+01 | 8.231e+00 | -3.794 | 0.000159 |
| ContinentEurope | -4.575e+01 | 8.247e+00 | -5.547 | 3.91e-08 |
| ContinentNorth America | -2.436e+01 | 8.528e+00 | -2.856 | 0.004396 |
| ContinentOceania | -2.360e+01 | 1.243e+01 | -1.899 | 0.057911 |
| ContinentSouth America | -2.418e+01 | 9.391e+00 | -2.575 | 0.010205 |
| gdp | -3.188e-13 | 1.044e-13 | -3.054 | 0.002329 |
| Length | 1.223e+00 | 3.262e-02 | 37.474 | $< 2e^{-16}$ |
| TunnelPer | 1.571e-01 | 1.702e-02 | 9.230 | $< 2e^{-16}$ |

Because of the difficulties of predicting cost per kilometer, we pivoted our focus to total cost.

We started with an initial model that did not use continents. When looking at the initial model (real_cost $\sim$ Stations + gdp + RR. + TunnelPer) AIC, BIC, R squared values and residual plots, a square root transformation seemed to be the best option to increase Adjusted R Squared without sacrificing model assumptions.

AIC and BIC lowest for log transformed, second lowest for sqrt, R squared highest for square root, log residual plot looked bad, sqrt looked okay, non transformed looked the best.

| Model | BIC | AIC | Adj_R2 |
|---|---|---|---|
| <chr> | <dbl> | <dbl> | <dbl> |
| real_cost | 15891.234 | 15862.876 | 0.5384437 |
| real_cost_sq | 7323.561 | 7295.204 | 0.5806689 |
| real_cost_log | 1947.668 | 1919.310 | 0.5042758 |

Figure 4: Histograms of Cost per kilometer Log Transformation

Once we settled on a sqrt transformation, we added all the possible predictors in (still excluding continents), then removed RR using bidirectional stepwise selection, removed urban pop and tunnel percentage because they had high p-values and were insignificant predictors. Stations was removed because it's highly correlated with length and this resulted in the best adjusted R squared.

Added continents and boom

```
lm(formula = sqrt(real_cost) ~ Years + Continent + gdp + Length +
    TunnelPer, data = df_clean)

Residuals:
    Min      1Q  Median      3Q     Max
-81.953  -9.312  -0.947   7.429  83.764

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)            5.791e+01  8.365e+00   6.924 8.86e-12 ***
Years                  4.884e-01  2.064e-01   2.367 0.018175 *
ContinentAsia         -3.123e+01  8.231e+00  -3.794 0.000159 ***
ContinentEurope       -4.575e+01  8.247e+00  -5.547 3.91e-08 ***
ContinentNorth America -2.436e+01  8.528e+00  -2.856 0.004396 **
ContinentOceania      -2.360e+01  1.243e+01  -1.899 0.057911 .
ContinentSouth America -2.418e+01  9.391e+00  -2.575 0.010205 *
gdp                   -3.188e-13  1.044e-13  -3.054 0.002329 **
Length                 1.223e+00  3.262e-02  37.474  < 2e-16 ***
TunnelPer              1.571e-01  1.702e-02   9.230  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 16.26 on 824 degrees of freedom
Multiple R-squared:  0.704,     Adjusted R-squared:  0.7008
F-statistic: 217.8 on 9 and 824 DF,  p-value: < 2.2e-16
```

Figure 5: Total Cost Model Summary

We first tried to build a large model with as many predictors as possible. From here, we looked at the correlation between our responses and our possible predictors. What stood out was correlation between these pairs: real cost and stations, real cost and length, stations and length, and real cost and start year. So, we removed one of each pair based on what resulted in the best adjusted R squared. We then removed insignificant predictors based on their p-values. And finally used stepwise selection (AIC) to make our last adjustments.

Our final model had 824 degrees of freedom when we started with 1007 observations because of NAs that weren't able to be removed. We experimented with transformations and found that applying a square root to real cost was beneficial to the adjusted R squared and resulted in the best looking residual plots.

We were also interested in why projects were left incomplete, but imputing missing values would have required making assumptions that weren't necessarily reasonable. Because we did not study this binary response variable very closely, logistic regression was not needed. We did not study years to complete for a similar reason.

## 4.3 First Model Diagnostics

To assess how reliable our model was and if the assumptions of the model were satisfied, we looked at our residual plots, the QQ plot along with a Kolmogorov-Smirnov test, GVIF—not VIF since Continents was factored—and conducted Durbin-Watson and Moran's I tests for spatial and auto correlation.

```
                GVIF Df GVIF^(1/(2*Df))
Years       1.143925  1        1.069544
Start.year  1.210187  1        1.100085
Continent   1.499177  5        1.041323
TunnelPer   1.103475  1        1.050464
PPP.rate    1.283217  1        1.132792
```

Figure 6: GVIF values for cost per kilometer model

All GVIF values were smaller than 2, so we had no concern about multicollinearity.

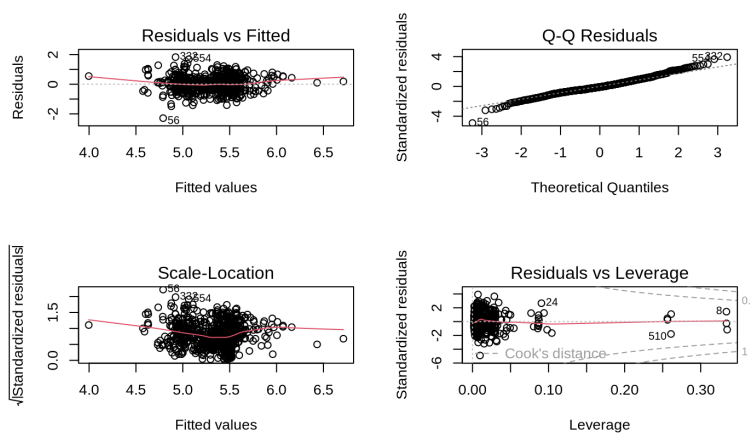


Figure 7: Cost per kilometer Diagnostic Plots

Looking at the residual plots, we see that most of our model assumptions are satisfied. There seems to be an even scattering of residuals around 0, with no evidence of heteroskedasticity, and the points on the QQ plot follow the linear line well, indicating normality of residuals.

Finally looking at autocorrelation and spatial correlation to attempt to satisfy our independence assumption, we conducted a Durbin-Watson and Moran's I test. The Durnin-Watson test indicated that there was evidence of autocorrelation, and the Moran's I test

indicated that there was no evidence of spatial correlation. We attempted various transformations, but these were unsuccessful in addressing autocorrelation. So, our independence assumption is not satisfied.

This mild violation of our regression assumptions may raise problems and reduce the accuracy of our model. It is likely that different transformations we did not consider, or some time-related structure that we did not consider in our model could further improve the model and satisfy all of our model assumptions.

## 4.4 Second Model Diagnostics

We approached model diagnostics for this model in the same manner as we did model 1. To assess how reliable our model was and if the assumptions of the model were satisfied, we looked at our residual plots, the QQ plot along with a Kolmogorov-Smirnov test, GVIF—not VIF since Continents was factored—and conducted Durbin-Watson and Moran's I tests for spatial and auto correlation.

```
               GVIF Df GVIF^(1/(2*Df))
Years     1.085705  1        1.041972
Continent 1.862839  5        1.064186
gdp       1.593384  1        1.262293
Length    1.204036  1        1.097286
TunnelPer 1.191187  1        1.091415
```

Figure 8: GVIF values for total cost model

All GVIF values were smaller than 2, so we had no concern about multicollinearity.

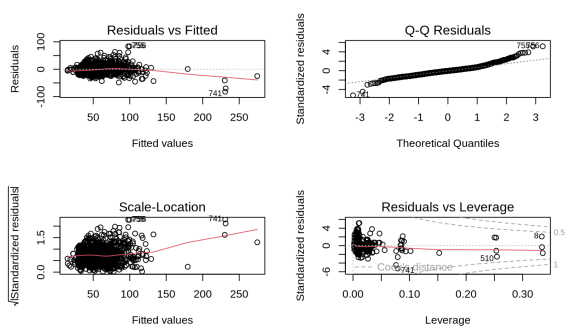


Figure 9: Total Cost Diagnostic Plots

Looking at the residual plots, we see some green and red flags. The residuals versus leverage plot shows us that there aren't any values with a Cook's distance above 0.5, meaning there aren't any highly influential points dominating the model. The QQ plot shows some deviations from normality in the tails and our Kolmogorov-Smirnov test failed, but since the sample size is so large, these are unlikely to be problematic. We can see that without the

higher fitted values, our residuals are approximately centered around 0 and there is relatively constant variance. However, after observing these outliers and researching these projects, these are valid data points and cannot be removed from our dataset.

Finally, we looked at autocorrelation and spatial correlation to attempt to satisfy our independence assumption. We conducted a Durbin-Watson and Moran's I test. Both of these resulted in small p-values and rejected their null hypotheses, indicating that there was evidence of spatial and auto correlation. Throughout the project, we attempted various transformations, and pulled in regional data such as gdp and urban population, to account for these. They were unsuccessful.

These mild violations of our regression assumptions may raise problems and reduce the accuracy of our model. It is likely that different transformations we did not consider, the addition of geographic or regional predictors, and other predictors could further improve the model and better satisfy the assumption.

# 5 Discussion and Conclusion

Summarize the main takeaways from your model in the context of the original problem. What do the coefficients mean in the real world?

Acknowledge the limitations of your study (e.g., potential omitted variable bias, non-representative sample) and suggest areas for future work.

# References

Levy, A., et al. Construction Costs of Urban Rail Projects Worldwide. New York University, 15 Jan. 2025, doi:10.58153/9wnjp-kez15. "Why Do Domestic Subway and Light Rail Projects Cost More and Take Longer to Construct than Similar Projects in Italy, Sweden, Turkey, Korea, Chile, and Spain?" Transit Costs Project, 6 Mar. 2026, transitcosts.com/. "GDP (Current US$)." World Bank Open Data, data.worldbank.org/indicator/NY.GDP.MKTP.CD?most_ Accessed 9 Mar. 2026. Brazil, Noli. "Lab 7: Spatial Autocorrelation." CRD 230: Spatial Methods in Community Research, 22 Feb. 2023, https://crd230.github.io/lab7.htmlx'x "Urban Population (% of Total Population)." World Bank Open Data, data.worldbank.org/indicator/SP.URB.T Accessed 10 Mar. 2026.

# A Appendix: R Code

Paste your analysis code here. If you prefer, you can use the `verbatim` environment to format it as raw text:

```
# Example R Code
model <- lm(Y ~ X1 + X2, data = mydata)
summary(model)
plot(model)
```